

RONIT JAIN

Backend Engineer · FinTech & ERP Systems · Node.js · Python · Azure · AI Pipelines

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Open to Relocation · Immediate Joiner

WHO I AM

For two years I was the backend engineer at Akssai ProjExel — not part of a backend team, but the backend engineer. FINAC, the ERP platform I owned, handles the financial operations of SME businesses: GST filing, invoice generation, bank reconciliation, payroll, payables, receivables. I designed every system from scratch, shipped to production, and kept 10+ modules running while the company scaled.

I didn't build CRUD apps. I built a tax compliance engine that automates GSTR-1 filing and integrates with government NIC APIs. I built a document intelligence pipeline that reads scanned invoices using Azure OCR and turns them into structured financial records. I built a bank statement parser that handles 15+ bank formats with fuzzy column matching and per-transaction confidence scoring. I resolved a production TCP port exhaustion that had 28,000 Redis connections deadlocking the entire infrastructure. That is the kind of work I do.

PROFESSIONAL EXPERIENCE

Software Engineer — Backend & AI Systems

Jul 2024 – Mar 2026

Akssai ProjExel · Jaipur, India

Sole backend engineer on FINAC — a live multi-tenant SaaS ERP platform serving SME financial operations across accounting, GST compliance, payroll, manufacturing, and CRM. Owned architecture, implementation, deployment, and production support across 10+ modules.

► Document Intelligence & OCR Pipeline

- Engineered end-to-end Azure Form Recognizer pipeline: scanned invoice → OCR extraction → structured record creation. Reduced manual invoice entry by ~80%; handles batch uploads of up to 1,000 invoices via Bull + Redis queue workers (10× faster than sequential).
- Designed per-field confidence scoring with configurable thresholds — stricter for amounts, looser for vendor names — surfacing extraction reliability for auditors; built exceptions dashboard for manual correction of low-confidence fields.
- Built idempotent batch reprocessing (same batch, no duplicate invoices), Unicode surrogate sanitization for downstream JSON safety, and default-branch fallback eliminating 100% of invoice creation failures from missing branch data.

► GST & Tax Compliance Automation

- Built complete GST token lifecycle: automated refresh every 6 hours, last_token_generated tracking, 30-day portal re-auth flow — eliminated manual intervention and API downtime for CA-facing workflows.
- Integrated GSTR-1 automated filing (payload mapping → GST schema → submission API → response parsing), GSTR-2A/2B reconciliation, E-Way Bill (NIC API), and E-Invoicing IRN (B2B Invoice Reference Number generation) — full compliance lifecycle in one platform.
- Built TDS Master module: deductee management, challan tracking, and TDS return aggregation for Chartered Accountant workflows.

► Multi-Format Bank Statement Parser

- Designed unified extract_transactions architecture (Python/Flask): Excel via openpyxl (handles merged cells), CSV via fuzzy header matching (no hardcoded column names), PDF via pdfplumber — one engine, 15+ bank formats, 1,000+ transactions per file.
- Reduced ledger mapping time ~75% — from 2–3 hours to 30–40 minutes. Confidence scoring per transaction flags low-confidence entries for auditor review. Integrated Pennyless API for account validation and metadata enrichment.

► PostgreSQL Performance Engineering

- Ran EXPLAIN ANALYZE across all slow reconciliation queries — found sequential scans on large multi-tenant tables. Added composite indexes on (tenant_id, financial_year, period) and partial indexes on soft-deleted rows (WHERE deleted_at IS NULL).
- Rewrote recursive chart-of-accounts queries (6–8 levels, ~50K records) with PostgreSQL recursive CTEs, eliminating N+1 application-level loops. Converted hot read paths to stored procedures. Cut query time from 2–5s to ~0.5s across the board.
- Enforced auth middleware across 300+ endpoints in multi-tenant schema; identified and patched SQL injection vulnerabilities in 3 search endpoints with parameterized queries.

► Production Reliability — Two War Stories

- TCP Port Exhaustion (Redis): Diagnosed 28,233 established connections in TIME_WAIT state via ss -s — consumed all ephemeral ports, halted all 8 Bull workers and 4 API services. Applied sysctl tuning (tcp_tw_reuse=1, tcp_fin_timeout=10, ip_local_port_range, somaxconn), persisted in /etc/sysctl.d/ — connections dropped 28,233 → 35.
- OOM Crash (Excel Reporting): 50K+ row reports were loading entire datasets into Node.js heap before writing — exhausted memory, crashed server. After 10 days of profiling and heap snapshots, rebuilt with ExcelJS streaming writer and paginated 5,000-row PostgreSQL reads. RAM stabilized permanently.

► Authentication, Integrations & Real-Time Systems

- Built JWT + CSRF double-submit cookie authentication for email/password and mobile OTP login flows — implemented end-to-end in the Node.js API layer.
- Implemented WebSocket server for real-time notification delivery; integrated Firebase FCM for mobile push notifications (token lifecycle, targeted dispatch); integrated Razorpay payment gateway (order creation, HMAC signature validation, webhook handling).
- Led migration of 100% Postman API collection to Bruno (Git-integrated): rewrote post-response scripts in Bruno syntax, pre-declared environment variables — zero regression, full team adoption.

Software Developer Intern

Jan 2024 – Jun 2024

Akssai ProjExel · Jaipur, India

- Built early GST API integrations and financial reporting modules; contributed across multiple FINAC backend modules. Promoted to full-time Software Engineer after 6 months based on performance.

PROJECTS

AI Document Processing System

Python · Flask · Pandas · pdfplumber · openpyxl

github.com/RonitRohil/Bank-Statement-Analyzer

- Multi-format bank statement parser processing 1,000+ transactions per file with dynamic header detection, per-transaction confidence scoring, and generalised metadata extraction (account number, IFSC, branch, holder name) across 15+ bank formats. Integrated Pennyless API for bank account and UPI verification.

TECHNICAL SKILLS

Languages: JavaScript (Node.js), Python, SQL

Backend Frameworks: Node.js, Express.js, Flask, FastAPI, REST APIs, WebSockets, GraphQL, Bull/Redis queue workers

Databases: PostgreSQL (stored procedures, recursive CTEs, EXPLAIN ANALYZE, composite & partial indexes, multi-tenant schemas), MongoDB (aggregation pipelines, CRM/HR modules), Redis (Bull queues), MySQL

AI & Data Engineering: Azure Form Recognizer (OCR), Pandas, pdfplumber, openpyxl, ExcelJS streaming, confidence scoring, fuzzy matching

Cloud & DevOps: Microsoft Azure (Blob Storage, Cognitive Services, VMs), Apache, PM2, Linux sysctl tuning

Integrations: Firebase FCM, Razorpay, GSTN APIs, E-Way Bill (NIC), E-Invoicing IRN, Pennyless API

Security & Architecture: JWT + CSRF double-submit cookie, parameterized queries / SQLi remediation, microservices, multi-tenant schema, async queue workers, streaming ETL

Dev Tools: Git, Postman, Bruno, Jira

EDUCATION

RECOGNITION & TRAINING

- Star Performer Award — Q1 2025, Akssai ProjExel: recognised for backend performance optimization, OCR pipeline delivery, and AI-based automation. Cited across 8+ manager check-ins as "the key person in our tech team", "acting as a senior developer", and "a role model for everyone in the tech team."

Course Completions (Udemy): [Azure Fundamentals \(AZ-900\)](#) · [AWS Cloud Practitioner \(CLF-C02\)](#) · [DevOps Beginners to Advanced](#)